PDF to Grouped Tree Structure Extractor

Transform unstructured PDF tables into standardised, hierarchical data structures with AI-powered intelligence

USER GUIDE

Understanding the Extraction Tool

What This Tool Accomplishes

The PDF to Grouped Tree Structure Extractor represents a sophisticated solution for data engineers and technical professionals who require precise extraction and validation of tabular data from PDF documents. This application leverages advanced table detection algorithms combined with AI-powered data mapping to transform complex, unstructured PDF content into standardised, hierarchical data structures.

The tool employs the Camelot library for initial table detection, supporting both lattice-based extraction for bordered tables and stream-based extraction for borderless content. Following extraction, SAP Cloud SDK for AI processes the data through generative models, intelligently mapping source fields to a consistent 29-column format whilst preserving data relationships and hierarchies.

Core Capabilities

- **Automated Table Detection:** Identifies and extracts tables using sophisticated pattern recognition
- **Intelligent Data Mapping:** AI models analyse content to map fields accurately to standardised columns
- **Hierarchical Structuring:** Transforms flat data into Solution/Tier/Server tree structures
- **Multi-Format Export:** Supports CSV, JSON, and XML output formats
- **Quality Validation:** Built-in verification ensures data completeness and accuracy
- **Selective Filtering:** Automatically excludes connectivity and migration services

Key Benefits and Use Cases

Automated Data Extraction

Eliminates manual data entry by automatically detecting and extracting tabular content from PDF documents. The tool processes multiple tables simultaneously, significantly reducing processing time whilst maintaining high accuracy levels. This automation frees technical resources to focus on analysis rather than data preparation.

AI-Powered Mapping Intelligence

Leverages advanced language models including GPT-4o and Claude 3.5 Sonnet to intelligently interpret source data structures. The AI understands context, handles variations in field naming, and accurately maps content to standardised columns. This intelligent processing adapts to different document formats without requiring manual configuration.

Standardised Output Format

Ensures consistency across all extractions by transforming diverse source formats into a uniform 29-column structure. This standardisation facilitates data comparison, validation workflows, and integration with downstream systems. Teams can rely on consistent field names, data types, and hierarchical relationships regardless of source document variations.

Flexible Export Options

Supports multiple output formats to meet varied workflow requirements. Export to CSV for spreadsheet analysis, JSON for programmatic access and API integration, or XML for validation and comparison workflows. Each format maintains data integrity whilst optimising for its intended use case.

STEP 1

PDF Upload and Configuration

Uploading Your Document

Begin by accessing the application's main interface and locating the file upload area, typically displayed as a dashed border region with upload instructions. Click this area to open your system's file browser, then navigate to and select your target PDF document.

Upon successful upload, the application validates the file format and displays confirmation including filename and file size. This validation ensures compatibility before proceeding to extraction. If upload fails, verify file permissions and ensure the document is a valid PDF format rather than a renamed image or document file.

Extraction Settings Configuration

The sidebar contains critical configuration options that determine extraction behaviour. The "Pages to extract" field accepts flexible input: specify "all" to process the entire document, use ranges like "1-5" for consecutive pages, or list specific pages as "1,3,5,7". Strategic page selection reduces processing time for large documents whilst ensuring all relevant content is captured.

Flavour Selection Guide

Lattice Mode: Optimal for tables with visible borders and gridlines. The algorithm detects line intersections to identify cell boundaries, providing high accuracy for structured tables commonly found in formal documents.

Stream Mode: Designed for borderless tables where cell boundaries are defined by spacing and alignment. This mode analyses text positioning to infer table structure, making it ideal for informal reports or documents without explicit borders.

Recommendation: Start with Lattice mode for most corporate documents. Switch to Stream only if initial extraction yields poor results.

Table Extraction Process

1

Initiation Phase

Click the "Extract & Structure Data" button to commence processing. The application enters Step 1 of 3, where the Camelot library analyses the specified PDF pages to identify table structures. Progress indicators display current status, and processing duration varies based on document complexity and page count.

2

Detection and Analysis

Camelot employs sophisticated algorithms to locate tables within the document. For Lattice mode, the system identifies intersection points of horizontal and vertical lines. Stream mode analyses text positioning, alignment patterns, and spacing to infer cell boundaries. Both methods generate candidate table regions for validation.

3

Extraction and Validation

Identified tables are extracted row by row, preserving cell content and structural relationships. The system validates each table by checking for consistent column counts, verifying data integrity, and ensuring readable content. Extraction summary displays table counts, locations, and dimensions for verification.

4

Results Presentation

Upon completion, the application presents an extraction summary detailing each discovered table. This summary includes page numbers, row counts, and column counts. Review these metrics carefully to ensure all expected tables were detected and that dimensions align with source document structure.

AI PROCESSING

Intelligent Data Structuring

AI Model Selection

The application automatically selects the optimal AI model from available deployments, prioritising advanced models such as GPT-4o, GPT-4, and Claude 3.5 Sonnet. This intelligent selection ensures maximum accuracy whilst optimising for response time and resource utilisation.

The chosen model receives extracted table data along with detailed instructions for mapping content to the standardised 29-column format. The prompt engineering includes field definitions, data type specifications, and examples to guide accurate interpretation.

Processing Pipeline

During Step 2 of 3, the AI model analyses extracted tables to identify field relationships, data types, and semantic meaning. It intelligently maps source columns to target fields, handles variations in naming conventions, and fills missing values based on context.

Processing typically requires 30-60 seconds depending on data volume and model availability. The system displays real-time progress indicators and handles timeouts gracefully by implementing retry logic with exponential backoff.

The 29-Column Standardised Structure

Understanding the standardised output format is crucial for effective data utilisation. The 29-column structure represents a comprehensive schema designed to capture all essential infrastructure configuration details whilst maintaining consistency across diverse source documents.

System Identity

Columns including amount, system_name, service, database, tiername, tier_type, and system_id establish unique system identification and categorisation. These fields enable grouping and filtering operations whilst providing essential context for infrastructure analysis.

Hardware Specifications

Fields such as ram_gib, cpus_physvirt, saps, no_of_add_hana_nodes, and no_of_standby_nodes capture compute resource allocations. These specifications support capacity planning, cost analysis, and performance modelling for infrastructure deployments.

Storage Configuration

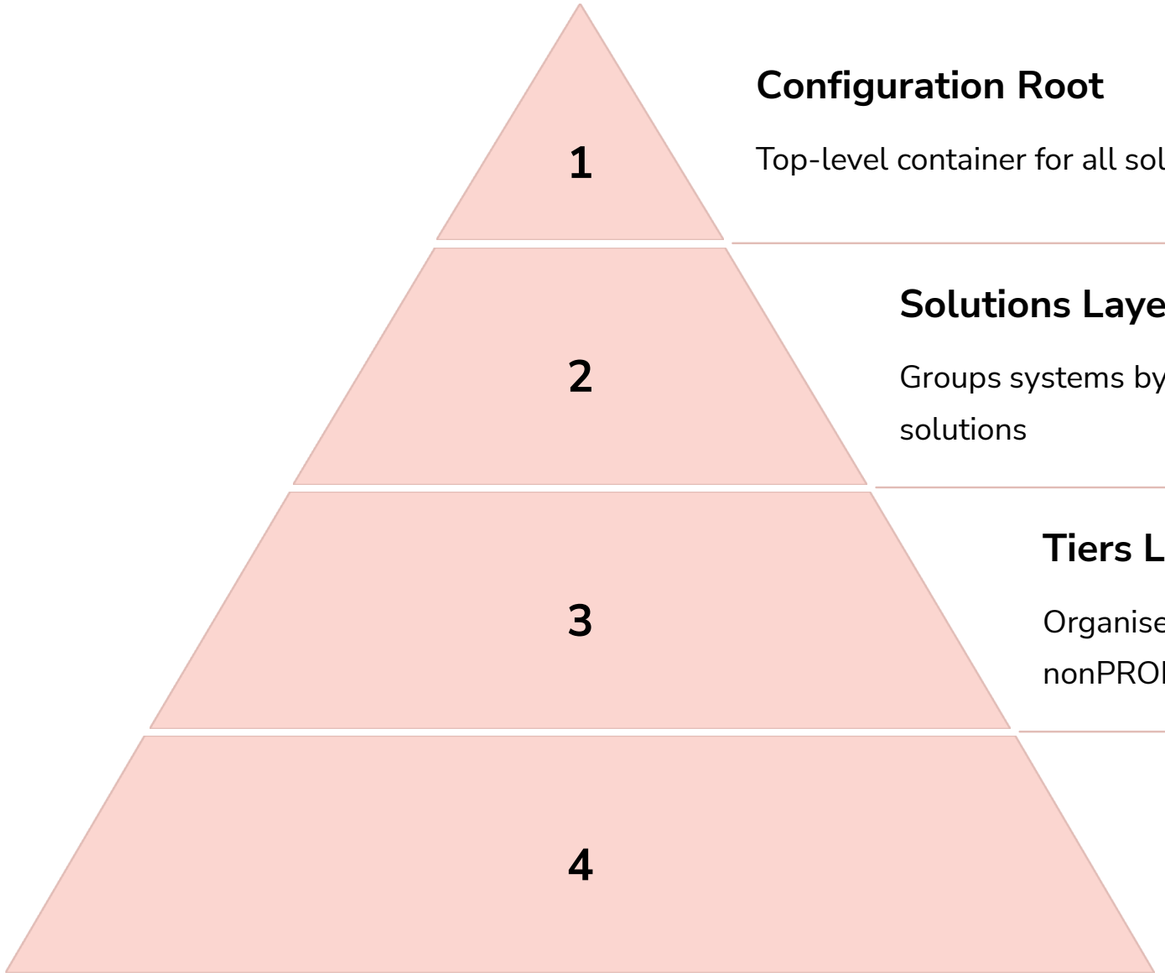
Multiple storage-related columns including tenant__user_data_size, amount_storage_1_gb, iops1, through_put1, amount_storage_2_gb, iops2, through_put2, and storage_information_1 provide comprehensive storage profiling. This granular detail supports storage architecture validation and cost optimisation.

Operational Parameters

Fields covering backup_class, os, sla, dr, add_hw_for_dr, pacemaker_included, add_requirements, phase, and server define operational characteristics and service level agreements. These parameters are essential for compliance verification and operational planning.

Grouped Tree Structure Transformation

Step 3 of 3 transforms the flat 29-column data into a hierarchical grouped tree structure, providing intuitive navigation and natural data organisation. This transformation groups related systems by service, tier, and server, reflecting typical infrastructure architecture patterns.



Configuration Root

Top-level container for all solutions

Solutions Layer

Groups systems by service name or service type, representing distinct application or infrastructure solutions

Tiers Layer

Organises systems within each solution by tier type, typically distinguishing PROD and nonPROD environments

Servers Layer

Individual server instances containing complete 29-column data for each configured system

OUTPUT FORMATS

Viewing and Interpreting Results

29-Column Table View

The default view presents structured data in a comprehensive table displaying all 29 columns across horizontally scrollable rows. This format provides immediate visibility into extracted content, supports sorting and filtering operations, and enables rapid data quality assessment. Use this view for initial validation and quick data review.

Raw XML (Step 1 Output)

This output represents the initial flat structure following AI mapping but before hierarchical transformation. Each row element contains all 29 columns as child elements, maintaining sequential ordering from the extraction process. This format is ideal for programmatic processing, comparison workflows, and integration with XML-based validation tools.

Grouped Tree View (Step 2 Output)

The hierarchical view organises data by Solution, Tier, and Server relationships, reflecting natural infrastructure architecture. Interactive expansion controls allow drilling down through the hierarchy to examine specific servers. This view supports filtering by solution name, tier type, and server name, with a comprehensive comparison table displaying filtered results. Use this view for architectural validation and relationship analysis.

JSON and CSV Formats

JSON output provides a structured, language-agnostic format ideal for API integration and programmatic access. CSV export delivers spreadsheet-compatible output for use in Excel, Google Sheets, or other data analysis tools. Both formats maintain complete data fidelity whilst optimising for their respective use cases.

Exporting Your Data

1

Select Appropriate Format

Choose the export format that best aligns with your intended workflow. CSV files integrate seamlessly with spreadsheet applications for manual analysis and reporting. JSON format suits programmatic processing, API integration, and automated validation pipelines. XML output supports comparison workflows and schema validation requirements.

2

Locate Download Controls

Each output view tab contains dedicated download buttons positioned for easy access. Navigate to the desired view—whether 29-Column Table, Raw XML, or Grouped Tree—and locate the download button, typically displayed with a downward arrow icon and descriptive text indicating the format.

3

Initiate Download

Click the download button to trigger file generation and browser download. Files are automatically named based on your source PDF filename with appropriate format extensions. For example, "contract_Q4_2024.pdf" generates "contract_Q4_2024_raw.xml", "contract_Q4_2024_grouped_tree.xml", and "contract_Q4_2024_29_column_output.csv".

4

Verify Download Integrity

After download completes, verify file integrity by checking file size (should not be zero bytes), confirming the file opens correctly in the appropriate application, and spot-checking data accuracy against the web interface display. Maintain downloaded files alongside source PDFs for audit trail and version control purposes.

PRACTICAL TRAINING

Hands-On Exercises

Exercise 1: Basic Extraction

Objective: Successfully extract and structure data from a simple PDF containing 1-2 tables using default settings.

Steps:

1. Upload a provided sample PDF with simple table structure
2. Configure settings using default values (all pages, Lattice mode)
3. Execute extraction and monitor progress through completion
4. Review 29-column table output for accuracy
5. Export results as CSV and verify in spreadsheet application

Success Criteria: All tables extracted successfully, data appears correctly in 29-column format, CSV export functions properly, and data matches source document.

Exercise 2: Advanced Navigation

Objective: Navigate complex hierarchical structures and apply filtering operations to isolate specific data subsets.

Steps:

1. Process a PDF containing multiple solutions and tiers
2. Navigate to grouped tree view and expand all hierarchy levels
3. Apply filters by solution name and tier type
4. Drill down to server details and verify all 29 columns
5. Export filtered results and compare with full dataset

Success Criteria: Can navigate tree structure efficiently, filters produce expected results, all data columns visible at server level, and filtered exports contain only selected data.

Exercise 3: Troubleshooting Challenge

This exercise develops critical troubleshooting skills by deliberately introducing common error scenarios. Practice identifying issues, diagnosing root causes, and implementing appropriate solutions.

1

Scenario 1: Table Detection Failure

Attempt to extract from a PDF without actual tables (contains only images or non-tabular text). Expected result: No tables found error. Practice resolution by identifying document limitations and attempting alternative extraction approaches. Learn to recognise when PDFs are unsuitable for automated extraction.

2

Scenario 2: Incorrect Flavour Selection

Process a borderless table using Lattice mode, resulting in poor extraction quality. Diagnose the issue by reviewing extraction summary and comparing with source document. Implement solution by switching to Stream mode and re-processing. Verify improved results.

3

Scenario 3: Connection Failure

Simulate credential issues by temporarily modifying .env file configuration. Attempt processing and observe connection failure messages. Practice diagnosing credential-related issues by reviewing error messages and testing connection. Restore proper configuration and verify successful connection.

4

Scenario 4: Data Quality Issues

Work with a PDF containing inconsistent table structures or data quality problems. Identify data completeness issues, mapping errors, and structural inconsistencies. Practice quality assessment techniques and learn to document issues for process improvement.

FAQ

Frequently Asked Questions: General

What is this application's primary purpose?

The application extracts structured data from PDF documents containing tables and transforms it into a standardised 29-column format organised in a hierarchical grouped tree structure. It's particularly valuable for processing contract documents, infrastructure quotes, and configuration specifications where consistent data structure facilitates validation and comparison.

What file formats are supported?

Currently, the application exclusively supports PDF format files containing actual tabular data with selectable text. Scanned images saved as PDFs require OCR processing before extraction can succeed. Document quality significantly impacts extraction success, with clear, well-defined tables yielding optimal results.

Are there file size limitations?

Whilst no hard-coded limit exists, very large PDFs increase processing time substantially. For optimal performance, consider extracting specific page ranges rather than processing entire large documents. Processing time scales approximately linearly with page count and table complexity.

Is internet connectivity required?

Yes, internet connectivity is essential because the application leverages SAP Cloud SDK for AI services requiring cloud access. All AI-powered data structuring operations depend on cloud-hosted language models. Local processing handles only initial PDF parsing and table detection phases.

FAQ: PDF Extraction Techniques

Lattice vs Stream Flavours

Question: When should I use Lattice versus Stream mode?

Answer: Lattice mode is optimal for tables with visible borders and gridlines, using line detection algorithms to identify cell boundaries. Stream mode suits borderless tables, using text alignment and spacing for structure detection. Start with Lattice for most corporate documents; switch to Stream only if detection fails.

Page Selection Syntax

Question: How do I specify which pages to extract?

Answer: Use flexible syntax in the page selection field: "all" processes entire documents, "1-5" handles page ranges, "1,3,5" specifies individual pages, and "1-3,5,7-10" combines ranges and individuals. Strategic selection improves efficiency for large documents.

Table Detection Failures

Question: What if no tables are detected?

Answer: Try switching flavours (Lattice to Stream or vice versa), verify the PDF contains actual tables rather than images, confirm page selection includes pages with tables, and ensure text is selectable rather than scanned imagery. Document quality significantly impacts detection success.

Scanned Document Handling

Question: Can I extract from scanned PDFs?

Answer: The application works best with PDFs containing actual text and table structures. Scanned PDFs containing only images require OCR processing before extraction can succeed. For optimal results, use PDFs with selectable text generated from native digital documents.

FAQ: AI Processing and Models

Model Selection and Usage

Q: Which AI model does the application use?

The application automatically selects from available models in your SAP Cloud SDK for AI deployment, preferring gpt-4o, gpt-4, gpt-4.1, Claude 3.5 Sonnet, Claude 4.5 Sonnet, or gpt-5 in that priority order. Automatic selection ensures optimal performance based on current availability.

Q: How long does AI processing typically take?

Processing duration varies based on table count, data volume, model response time, and network latency. Typical processing for moderate-sized documents ranges from 30-60 seconds. Large or complex documents may require additional time.

Processing Issues and Solutions

Q: What causes AI processing failures?

Common causes include connection errors indicating network or credential issues, timeouts suggesting excessive data volume or slow model response, invalid responses indicating model deployment problems, and mapping errors stemming from unusual source table structures.

Q: Can I customise the AI prompt?

The AI prompt is built into the application to ensure consistent 29-column mapping across all extractions. Customisation would require code modifications and could impact mapping consistency. The standardised prompt has been optimised through extensive testing.

FAQ: Understanding Data Output

01

29-Column Structure Definition

The standardised output comprises system information fields (amount, system_name, service, database, tiername, tier_type, system_id), hardware specifications (ram_gib, cpus_physvirt, saps, no_of_add_hana_nodes, no_of_standby_nodes), storage configuration (tenant__user_data_size, amount_storage_1_gb, iops1, through_put1, amount_storage_2_gb, iops2, through_put2, storage_information_1), and operational parameters (backup_class, os, sla, dr, add_hw_for_dr, pacemaker_included, add_requirements, phase, server).

02

Raw XML vs Grouped Tree XML

Raw XML presents a flat structure with all rows at the same level, where each row element contains all 29 columns. This format suits validation workflows and programmatic processing. Grouped Tree XML provides hierarchical organisation by Solution, Tier, and Server, grouping related data naturally for architectural analysis and comparison operations.

03

Automatic Filtering Logic

The application automatically excludes connectivity services (VPN, Direct Connect, VPC Peering), migration services, and supplementary offerings. This filtering focuses output on core infrastructure components requiring detailed configuration tracking, reducing noise and simplifying downstream analysis.

04

Export Format Selection

Three formats serve distinct purposes: CSV for spreadsheet applications and manual analysis, JSON for programmatic access and API integration, and XML for validation workflows and structural comparison. Select formats based on your specific workflow requirements and downstream system compatibility.

CASE STUDY

Real-World Walkthrough: QT_real_sample.pdf

This comprehensive case study demonstrates the complete extraction workflow using an actual quote document. Follow this walkthrough to understand practical application of all concepts covered in previous sections.

Document Overview

QT_real_sample.pdf represents a typical infrastructure quote document containing multiple tables with system configurations, hardware specifications, storage allocations, and operational parameters. The document spans several pages with tables featuring both bordered and mixed formatting.

This case study provides realistic expectations for processing time, extraction quality, and output characteristics. Following this walkthrough helps you understand what constitutes successful extraction versus situations requiring intervention or adjustment.

Learning Objectives

- Experience complete end-to-end workflow execution
- Understand realistic processing times and outcomes
- Practice validation techniques on actual output
- Learn to identify and address quality issues
- Develop confidence with real-world scenarios

Case Study: Preparation Phase

Pre-Processing Verification

Before commencing processing, verify SAP Cloud SDK for AI connection by clicking "Test Connection" in the sidebar. Ensure you receive successful connection confirmation with deployment details. This verification prevents mid-processing failures due to credential or connectivity issues. Also confirm sufficient disk space for output file generation and stable network connectivity for AI processing.

Document Assessment

Locate QT_real_sample.pdf in your project directory and verify file integrity by checking file size and confirming the document opens correctly in a PDF viewer. Quickly scan the document to identify pages containing tables, note table structure characteristics (bordered vs borderless), and estimate processing scope based on document complexity.

Workspace Preparation

Open the application in your preferred web browser with adequate screen resolution for viewing detailed tables. Ensure the sidebar is expanded for easy access to configuration options. Clear any previous session data if the application retains state. Position your PDF viewer alongside the application window for convenient comparison during validation.

Expected Setup Duration: 2-3 minutes for initial preparation

Case Study: Document Upload

Upload Execution

Navigate to the main application area and locate the file uploader component, typically displayed with a dashed border and upload icon. Click this area to open your system's file browser, navigate to your project directory, select QT_real_sample.pdf, and click Open to initiate upload.

The application validates the file format and displays upload confirmation including filename and file size. Typical upload for moderate-sized PDFs completes within seconds, though duration varies with file size and network speed.

Upload Verification

Confirm successful upload by checking for the success message "✅ File uploaded" in the interface. Verify the displayed filename matches your selected document: "QT_real_sample.pdf". Check that file size is displayed and appears reasonable for the document.

If upload fails, common causes include insufficient file permissions, network connectivity interruptions, or file format issues. Verify file integrity by opening in a PDF viewer and retry upload after addressing any identified issues.

Pro Tip: For very large documents, consider testing extraction on a subset of pages first (e.g., pages 1-3) to verify configuration and table detection quality before processing the entire document. This approach saves time when adjustments are needed.

Case Study: Extraction Configuration

Page Selection Strategy

For QT_real_sample.pdf, configure page selection in the sidebar's "PDF Extraction" section. For comprehensive processing, enter "all" in the "Pages to extract" field to capture all tables throughout the document. This ensures no data is inadvertently excluded.

Alternative approach: If processing time is a concern or you're conducting initial quality testing, specify a subset such as "1-5" to process only the first five pages. This approach allows rapid validation before committing to full document processing.

Flavour Selection Rationale

For QT_real_sample.pdf, which typically contains tables with defined borders as found in formal quote documents, select "lattice" from the "Camelot Flavor" dropdown. Lattice mode excels at detecting tables with visible gridlines and borders, using line intersection algorithms for precise cell boundary identification.

If initial extraction yields poor results (no tables detected or incomplete extraction), switch to "stream" mode and re-process. Stream mode works better for borderless tables where cell boundaries are defined by spacing and alignment rather than visible lines.

Recommended Configuration for QT_real_sample.pdf:

- Pages to extract: all
- Camelot Flavor: lattice

Case Study: Table Extraction Results

Initiating Extraction

Click the "🔄 Extract & Structure Data" button to commence processing. The application immediately enters Step 1 of 3, displaying progress indicators and status messages. Monitor the interface for "📊 Step 1/2: Extracting tables from PDF using Camelot..." message.

For QT_real_sample.pdf, extraction typically completes within 10-30 seconds depending on document size and system performance. The application displays a progress spinner during processing. Avoid closing the browser or navigating away during this phase.

Expected Extraction Results

Upon completion, expect to see a success message: "✅ Extracted 3-5 table(s) from PDF" (exact count depends on document content). Expand the "📋 Table Extraction Summary" to review details for each detected table.

Each table entry displays page number, row count, and column count. For QT_real_sample.pdf, typical results show tables on pages 1-4 with varying dimensions reflecting different data presentations in the quote document.

3-5

Tables Detected

Typical extraction count for quote documents of this type

20-30

Total Rows

Combined row count across all detected tables

6-10

Columns per Table

Average column count in source tables

Case Study: Grouped Tree Generation

Automatic Transformation

Following AI structuring, the application automatically proceeds to Step 3 of 3: grouped tree generation. This transformation happens rapidly (typically within seconds) as the system reorganises flat 29-column data into hierarchical Solution-Tier-Server relationships.

Watch for the status message "🌳 Step 3/3: Generating grouped tree structure..." followed quickly by completion confirmation. The transformation algorithm analyses system relationships, groups by service name (Solution), organises by tier type (PROD/nonPROD), and nests individual servers with complete data.

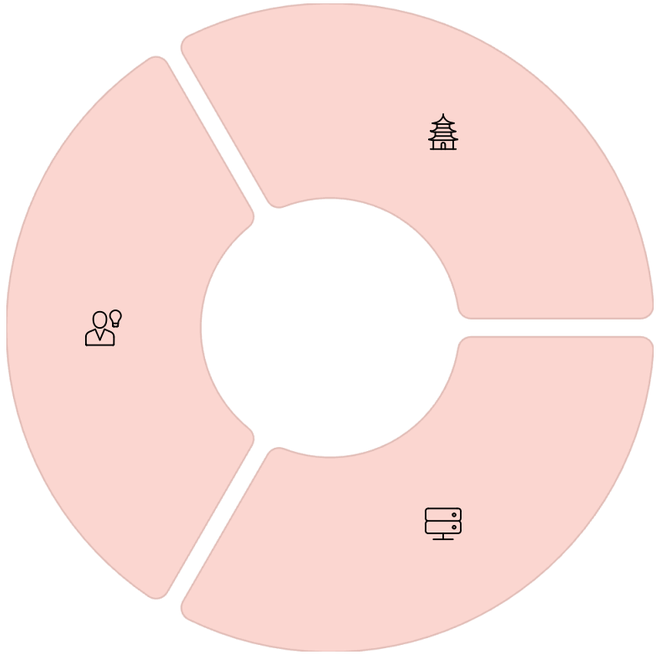
Expected Tree Metrics

For QT_real_sample.pdf, typical results show 2-4 unique Solutions representing distinct services or applications, 4-8 Tiers reflecting PROD and nonPROD environments across solutions, and 10-20 Servers representing individual system instances with complete 29-column configurations.

The success message displays these counts: "✅ Generated grouped tree: X Solution(s), Y Tier(s), Z Server(s)". Review these metrics to ensure they align with expectations based on the source document content.

Solutions Layer

2-4 unique services representing distinct applications or infrastructure solutions in the quote



Tiers Layer

4-8 environment tiers distinguishing PROD and nonPROD deployments across solutions

Servers Layer

10-20 individual server instances each containing complete 29-column configuration details

Case Study: Navigating the Grouped Tree

01

Initial Tree View

Click the "🌳 Grouped Tree (Step 2)" tab to access the hierarchical view. The interface displays summary metrics showing total counts for Solutions, Tiers, and Servers. These counts provide an immediate overview of data organisation and help verify expected structure.

02

Expanding Solutions

Click the expander icon next to "Solutions" to reveal individual Solution nodes. Each Solution represents a distinct service or application from the quote document. Solution names derive from the "service" field in the 29-column data, grouping related systems together.

03

Drilling to Tiers

Expand individual Solution nodes to reveal their Tier children. Tiers typically include both PROD and nonPROD categories, reflecting production and non-production environments. The tier organisation enables filtering and analysis by environment type, critical for validation and comparison workflows.

04

Examining Server Details

Expand individual Tier nodes to reveal Server children. Each Server represents a single system configuration with all 29 columns displayed in a structured table format. This detailed view enables verification of complete data preservation through the transformation process. Scroll through the server table to examine all field values.

Case Study: Using Filtering Features

Solution Filtering

Below the tree structure, locate the "Complete Server Comparison Table" section containing filtering controls. The "Filter by Solution" dropdown lists all detected solutions from the grouped tree. Selecting a solution filters the comparison table to display only servers belonging to that solution.

This filtering capability proves valuable when validating specific services or comparing configurations across solutions. The filtered view maintains all 29 columns whilst limiting rows to the selected solution's servers.

Tier Type Filtering

Use the "Filter by Tier Type" dropdown to isolate PROD or nonPROD servers. This filtering enables focused analysis of production configurations separately from development or testing environments. Combined with solution filtering, you can examine specific environment subsets such as all production servers for a particular application.

Server Name Search

The "Search Server Name" text field enables free-text searching within server names. Enter partial matches to quickly locate specific systems. This search combines with solution and tier filters, allowing precise isolation of target configurations.

For QT_real_sample.pdf, searching for "HANA" might isolate all database servers, whilst "PROD" in combination with solution filtering could identify all production instances of a specific application.

Case Study: Comprehensive Data Validation

Row Count Verification

Compare the total row count in the 29-column table against your manual count of systems in QT_real_sample.pdf. Discrepancies indicate potential extraction issues such as missing tables, filtered services, or extraction configuration problems. The summary section displays total rows and rows with data for quick verification.

System Name Accuracy

Verify that all system names from the source document appear in the structured output. Use the grouped tree view to navigate through solutions and tiers, checking that expected systems are present. Missing systems suggest detection or mapping issues requiring investigation.

Hardware Specification Validation

Spot-check hardware specifications (RAM, CPU, SAPs) against source document values. Select several representative systems and verify exact matches. Discrepancies in numeric values indicate AI mapping errors or source table ambiguity. Storage allocations should also match quoted values precisely.

Tier Classification Verification

Confirm that tier_type values correctly reflect production versus non-production status as indicated in the source document. Misclassifications could impact downstream workflows that treat PROD and nonPROD differently. Review tier assignments across multiple systems to ensure consistency.

Case Study: Exporting Results

1

CSV Export for Spreadsheet Analysis

Navigate to the "📊 29-Column Table" tab and click the "⬇️ Download CSV" button. The file downloads as "QT_real_sample_29_column_output.csv" and opens directly in Excel, Google Sheets, or similar applications. This format enables sorting, filtering, pivot tables, and charting for detailed analysis.

2

Raw XML Export

In the "📄 Raw XML (Step 1)" tab, click "⬇️ Download Raw XML" to save the flat XML structure as "QT_real_sample_raw.xml". This format suits validation tools expecting flat structures, integration with XML-processing pipelines, and comparison against other quote versions using XML diff tools.

3

Grouped Tree XML Export

Navigate to the "🌳 Grouped Tree (Step 2)" tab, scroll to the bottom, and click "⬇️ Download Grouped Tree XML" to save as "QT_real_sample_grouped_tree.xml". This hierarchical format enables architectural validation, comparison of solution structures, and integration with systems expecting nested XML.

4

Export Verification

After downloading all formats, verify each file opens correctly in appropriate applications, check file sizes are non-zero and reasonable, confirm data matches what appeared in the web interface, and ensure filenames follow the expected pattern. Store all exports alongside the source PDF for complete audit trail.

Case Study: Expected Results Summary

Quantitative Outcomes

Based on typical quote documents similar to QT_real_sample.pdf, expect these approximate metrics:

- **Tables Extracted:** 3-5 distinct tables from various document pages
- **Rows Structured:** 15-30 individual system configurations
- **Solutions:** 2-4 unique services or applications
- **Tiers:** 4-8 environment tiers across all solutions
- **Servers:** 10-20 complete server configurations

Variations from these ranges don't necessarily indicate problems—document content drives actual counts. However, significant deviations warrant investigation.

Qualitative Assessment

Data Quality: High-quality extractions show minimal empty critical fields, accurate hardware specifications, consistent naming, and proper tier classification.

Grouping Accuracy: Solutions should reflect actual services from the source document, tiers should correctly distinguish PROD/nonPROD, and servers should nest under appropriate parents.

Completeness: All relevant systems from the quote document should appear in output. Filtered connectivity and migration services are expected exclusions.

Measuring Success and ROI

80%

Time Reduction

Typical time savings versus manual data entry for infrastructure quotes

95%

Accuracy Improvement

Reduction in manual transcription errors through automated extraction

100%

Format Standardisation

Consistency achieved across all extracted quote data regardless of source

60%

Validation Efficiency

Improvement in quote comparison and validation workflow throughput

Track quantifiable metrics demonstrating tool value including time savings per extraction compared to manual approaches, error reduction in data transcription and validation, processing volume enabled by automation, and integration efficiency gains in downstream workflows. These metrics justify continued investment and support expansion to additional use cases.